

MUHAMMAD WALEED TARIQ

Software Engineer

Manchester | waleedtariq.com |  [GitHub](#) |  [LinkedIn](#)

Skills

Programming & Frameworks: Java, Python, JavaScript, Spring Boot, React, Gradle

DevOps & Tools: IntelliJ, AWS, K6, Jmeter, Docker, Kubernetes, Jenkins, Kafka, ActiveMQ, Tomcat, Nginx

Databases & Monitoring: MSSQL, PostgreSQL, MongoDB, Prometheus, Grafana, New Relic

Methodologies & Practices: System Design, Agile, Scrum, Pair programming, Jira, Confluence

Experience

Software Engineer

Sainsbury's, Manchester

March 2025 – Present

- Part of Digital Fulfilment team, developing critical backend services that calculate accurate delivery timelines for customer orders.
- Designing and developing a new order reservation system, enabling advanced inventory management and improved customer experience through real-time stock availability.
- Building event-driven microservices using Spring Boot and Apache Kafka, handling high-volume data streams for real-time order processing and inventory updates.
- Architecting solutions using diverse data stores (PostgreSQL, MongoDB, DynamoDB) to optimize data access and performance.
- Managing deployments and infrastructure on AWS using Kubernetes, ensuring high availability and scalability of services.
- Collaborating with product teams and stakeholders to design and implement solutions that improve order fulfillment accuracy and efficiency.

Software Engineer I

THG Ingenuity, Manchester

July 2022 – March 2025

- Developed and maintained order-processing microservices handling order allocation and shipment generation using Java, Spring Boot, ActiveMQ, Tomcat, Docker, and MSSQL
- Led Black Friday 2024 performance optimization, improved SQL efficiency, decoupled dependencies, monitored metrics, and implemented load testing to boost system stability.
- Built CI/CD pipelines with Jenkins for Kubernetes and Linux-based deployments; collaborated across teams to onboard new clients while maintaining backward compatibility
- Resolved high-priority client incidents, providing concise impact analysis and effective production fixes

Machine Learning Researcher

Lancaster University, Lancaster

June 2022 – September 2022

- Extended final-year project to fine-tune and evaluate YOLOv3/v5 models for landfill detection from satellite and drone imagery using Google Cloud
- I collected a training dataset containing 2,000 high quality images through Google Earth & open-source repositories.

Personal Projects

Spring Boot Learning Kit (★ 212+)

- Developed an enterprise-grade learning project featuring 11 hands-on tasks for new learners.
- Covers topics from implementing REST APIs to migrating Database Schemas, technologies like Apache Camel, ActiveMQ, RabbitMQ and PostgreSQL, Prometheus & Grafana.
- **Repository:** [GitHub](#)

Nimbus

- Personal Cloud Storage webapp built with JS, React, Tailwind CSS.
- Allows file upload, download & directory navigation.
- Backend application is built with Java Spring Boot which uses Amazon S3 to store files.
- **Repository:** [GitHub](#)

Fly Drone with an Xbox Controller

- DJI Tello drone can be controlled via smart phone, but it does not have built-in support for a controller, so I decided to create a script using Tello SDK & API which lets you fly the drone with an Xbox Controller.
- The script can also stream the live video feed from the drone's camera to your device.
- **Repository:** [GitHub](#)

Education

BSc Hons In Computer Science

Lancaster University, Lancaster

Sept 2019 - June 2022

Grade: 2:1

Relevant Coursework

- Developed a distributed auctioning system using “Java RMI” allowing sellers and buyers to create and place bids on items. Backend used active replication to ensure data reliability and availability.
- Lead a team in development of an OpenGL based 2D platform fighter game using SFML in Java.
- **Dissertation:** Worked with a machine learning model to process drones and satellite imagery to evaluate littering across UK.

Foundation Year in Computer Science

University of Manchester, Manchester

Sept 2018 - June 2019

Grade: 1st Class

Relevant Coursework

Mathematics with Mechanics, Chemistry, Physics, Foundation Year Project

Languages

- **English**
Fluent
- **Urdu**
Native

Certifications

- Full UK Driving License